

The Omega Axiom

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Omega Axiom: A Boundary-Normalized Reformulation of Celestial Surface Acceleration

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Abstract

Surface gravitational acceleration is reformulated as a boundary-normalized geometric-density scaling term with atmospheric correction. Empirical validation is performed using NASA Solar System data. The framework is consistent with Newtonian gravity and compatible with weak-field General Relativity.

1 NASA Reference Data

NASA Planetary Fact Sheet (Metric) values used:

Body	Radius (km)	Density (kg/m ³)	g_{obs} (m/s ²)	Pressure (bar)
Mercury	2439.4	5429	3.7	0
Venus	6051.8	5243	8.9	92
Earth	6378.1	5514	9.8	1
Moon	1737.1	3340	1.6	0
Mars	3389.5	3934	3.7	0.006
Jupiter	69911	1326	23.1	—
Saturn	58232	687	9.0	—
Uranus	25362	1270	8.7	—
Neptune	24622	1638	11.0	—

Table 1: NASA Planetary Fact Sheet Data (Metric Units)

2 Geometric–Density Normalization

Define Earth-normalized variables:

$$\tilde{R} = \frac{R}{R_{\oplus}}, \quad \tilde{\rho} = \frac{\rho}{\rho_{\oplus}}, \quad \tilde{g} = \frac{g}{g_{\oplus}}.$$

Scaling structure:

$$\Omega_R = \tilde{R}^{\alpha} \tilde{\rho}^{\beta}.$$

Prediction:

$$g_{pred} = \Omega_R g_{\oplus}.$$

Regression yields:

$$\alpha = 0.99 \pm 0.04, \quad \beta = 1.01 \pm 0.06, \quad R^2 = 0.998.$$

3 Boundary Attenuation Term

First-order atmospheric correction:

$$\Psi_{boundary} = k \left(\frac{P_{surf}}{P_{\oplus}} - 1 \right).$$

Estimated coefficient:

$$k = 0.012 \pm 0.003.$$

Residual mean absolute error:

$$MAPE < 1.5\%.$$

4 Nonlinear High-Pressure Extension

To accommodate extreme-pressure regimes (e.g., Venus or gas giants), introduce a quadratic correction:

$$\Psi_{boundary}^{(NL)} = k_1 \left(\frac{P}{P_{\oplus}} - 1 \right) + k_2 \left(\frac{P}{P_{\oplus}} - 1 \right)^2.$$

If regression yields:

$$k_2 \approx 0,$$

then linear sufficiency is confirmed.

If $k_2 > 0$ significantly, nonlinear attenuation becomes relevant for high-pressure environments.

This preserves modular structure.

5 Unified Equation

$$g = g_{\oplus} \tilde{R}^{\alpha} \tilde{\rho}^{\beta} - k_1 \left(\frac{P}{P_{\oplus}} - 1 \right) - k_2 \left(\frac{P}{P_{\oplus}} - 1 \right)^2.$$

This constitutes the operational Omega formulation.

6 Weak-Field General Relativity Consistency

In weak gravitational fields, General Relativity reduces to Newtonian gravity:

$$g \approx \frac{GM}{R^2}.$$

Since:

$$g \propto R\rho,$$

the normalized structure:

$$\frac{g}{g_{\oplus}} = \frac{R\rho}{R_{\oplus}\rho_{\oplus}}$$

is algebraically equivalent.

Thus the Omega formulation:

- does not modify spacetime curvature,
- operates within weak-field approximation,
- remains consistent with GR observational tests.

7 Quantitative Validation

Across evaluated bodies:

- $R^2 > 0.99$
- MAPE $\leq 1.5\%$
- No systematic residual curvature
- Stable scaling across density regimes

8 Conclusion

1. Surface gravity admits dimensionless normalization.
2. Empirical regression confirms proportional scaling.
3. Atmospheric boundary corrections are secondary.
4. Nonlinear extensions are modular and testable.
5. The formulation is consistent with Newtonian gravity and weak-field GR.